Ogbonna Arinze

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SUMMARY

Versatile Senior Full-Stack Engineer with 6+ years of experience building high-impact, scalable web applications using TypeScript, React, Next.js, and Node.js across performance-critical environments. Proven expertise in architecting end-to-end features, from database design and backend services to responsive, accessible frontends, with deep experience in GraphQL, SQL/NoSQL databases, and AWS cloud services. Passionate about building reusable frameworks, internal tooling, and shared utilities that elevate developer experience across engineering teams, while mentoring engineers and driving best practices in code quality, testing, and CI/CD.

CORE COMPETENCIES

- **Full-Stack Architecture:** End-to-end feature design, scalable component systems, micro-frontends, monorepos (Turborepo, Nx), SSR/SSG/ISR strategies, and reusable shared frameworks.
- **Programming Languages:** TypeScript, modern JavaScript (ESNext), Node.js, Python, and PHP.
- **Frameworks & Tools:** React, Next.js, Vue.js, Nest.js, Tailwind CSS, Redux, React Query, Storybook, and Webpack/Vite/Rollup for custom build pipelines.
- **Backend & Data:** Node.js services, GraphQL (Apollo Federation), REST APIs, PostgreSQL, MongoDB, Redis, DynamoDB, Prisma/Drizzle ORM, and Kafka event streaming.
- **Cloud & Infrastructure:** AWS (EC2, S3, CloudFront, DynamoDB), Docker, Kubernetes, Vercel, edge rendering, and CI/CD automation (GitHub Actions).
- **Quality & Reliability:** Jest, Cypress, Playwright, automated testing pipelines, CI-driven QA, TDD/BDD, and release strategies.
- **Observability & Monitoring:** Sentry, Datadog RUM, LogRocket, Lighthouse, Chrome DevTools, and synthetic monitoring.
- **Leadership & Methodologies:** Technical direction, architecture reviews, mentorship, Agile/Scrum, and cross-team collaboration.

WORK EXPERIENCE

Roche Genentech — San Francisco, CA (Remote)

Senior Full-Stack Engineer — *May 2023 - Present*

- Led the design and implementation of scalable, end-to-end features for a clinical trial data platform, architecting both the **Node.js/TypeScript** backend services and the **React** frontend, ensuring high performance and reliability across the full stack.
- Architected a reusable **design system** using **React** and **Radix UI** that served as a shared framework across 5+ product teams, improving UI consistency, accessibility (WCAG), and reducing frontend development time by 40%.
- Designed and delivered **GraphQL APIs** with efficient resolver patterns and schema stitching, enabling flexible data fetching for complex clinical trial dashboards while reducing over-fetching and improving client-side performance.
- Built and optimized backend services using **Node.js** and **TypeScript**, implementing RESTful and **GraphQL** endpoints with robust error handling, input validation, and structured logging for production-grade reliability.
- Engineered **JWT-based authentication** and **role-based access control (RBAC)** in a **Next.js** application with **TypeScript** and **React Query**, implementing HTTP-only cookie token handling and collaborating with backend teams to enforce permission-based UI rendering and API authorization.
- Scaled state management in a large **Next.js** application by integrating **Redux** for complex client state and **React Query** for cached API data, streamlining data flow, cutting redundant re-renders, and accelerating developer onboarding through consistent, maintainable patterns.
- Optimized application performance in a **React** and **TypeScript**-based SPA by implementing lazy loading, code splitting, dynamic imports, and image optimization, reducing load times significantly. Enhanced responsiveness

with **Radix UI** layouts, memoization, and efficient **CDN asset delivery** via **AWS CloudFront/S3**.

- Designed and maintained **PostgreSQL** database schemas and query patterns using **Prisma ORM**, optimizing complex queries for clinical data retrieval and implementing migrations for evolving data models in a production environment.
- Transformed a legacy **Vue.js** web app into a **React Chrome extension**, re-engineering both the frontend and the underlying **Node.js** service layer to optimize workflows for clinical site integrations and enhance user adoption.
- Increased release reliability by integrating **Playwright** end-to-end tests and **Jest** unit tests into the **CI/CD pipeline** via **GitHub Actions**. Automated test reporting with **Allure Reports** and configured **Slack** notifications for real-time build quality visibility.
- Created reusable utilities, shared libraries, and reference implementations in **TypeScript** that were adopted across multiple product teams, standardizing patterns for API integration, error handling, and data validation.
- Mentored junior and mid-level engineers through code reviews, pair programming, and architecture discussions, fostering a culture of technical excellence, collaboration, and continuous improvement across the team.

DESCASIO LTD Lagos, Nigeria

Senior Full-Stack Engineer *May 2021 - Apr 2022*

- Owned the core full-stack architecture for a data-heavy **React** dashboard backed by **Node.js** services, implementing virtualization, efficient state management, and optimized **SQL** queries to handle large datasets seamlessly.
- Architected and maintained **Node.js/TypeScript** backend services with **RESTful** and **GraphQL** APIs, designing scalable data access layers using **PostgreSQL** and **MongoDB** to support complex reporting and analytics features.
- Optimized frontend performance by identifying and resolving render bottlenecks using **React Profiler** and **Lighthouse Audits**, reducing bundle size by 30-40% through **rollup-plugin-visualizer** and **webpack-bundle-analyzer**, achieving a 35% faster load time across devices.
- Designed and built internal admin tooling and CMS solutions using **React**, **TypeScript**, and **Node.js**, enabling non-technical stakeholders to manage content and configurations without engineering support.
- Containerized application services using **Docker** and **Docker Compose**, streamlining local development workflows and enabling consistent deployment across staging and production environments on **AWS**.
- Implemented **Redis**-based caching strategies for frequently accessed API endpoints, reducing database load by 50% and improving response times for data-intensive dashboard views.
- Designed and delivered reusable utilities, shared component libraries, and reference implementations used across 3+ product teams, standardizing frontend patterns and accelerating feature delivery.
- Established engineering standards for code quality, testing, and deployment across the team using **ESLint**, **Prettier**, **Husky**, **Lint-staged**, and **GitHub Actions**, reducing errors and accelerating **CI/CD** processes.
- Led standardization efforts for the engineering team, creating technical guidelines, documentation, and best practices for **TypeScript**, **React**, and **Node.js** development that improved code maintainability and developer experience.
- Mentored junior and mid-level engineers, conducting regular code reviews and knowledge-sharing sessions, fostering a culture of high-quality, maintainable code and consistent design across projects.

EDUCATION

University of Nigeria Nsukka, Enugu

BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*